

Mary Hiffa

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EDUCATION

Rensselaer Polytechnic Institute (RPI)

B.S. in Computer Science and Mathematics, Minor in Music, *magna cum laude*. GPA: 3.79/4.0

Troy, NY

MAY 2025

RELEVANT COURSEWORK

Machine Learning from Data, Numerical Computing, Numerical Linear Algebra with Applications, Graph Theory, Number Theory, Operating Systems, Data Structures

SKILLS

Programming Proficiencies: Python, C, C++, HTML, CSS, JavaScript, Typescript, Java, SQL, Matlab, NumPy, Pandas, PyTorch, JUnit, Node.js, Express, React, Git, Linux, Algorithm Design, Software Design, UI/UX

Other: Data Analysis, Mathematical Optimization, English Writing & Editing

WORK EXPERIENCE

Troy Tutors

Computer Science and Mathematics Tutor

Troy, NY

JUN 2023 – PRESENT

- Tutor high school and undergraduate students in mathematics and computer science. Over 70% of sessions involve recurring students who report marked grade improvement.
- Focused on aiding students in RPI coursework which relates discrete math to computer science as well as RPI coursework in Java-based object-oriented programming.

Grill 41

Sous Chef/Server

Hanover, MA

MAY 2017 – JUL 2024

- Worked as sous chef, grill chef, and head server for family-owned seasonal catering business.

LEADERSHIP

Rensselaer Center for Open Source

Project Lead – DSR Website

Troy, NY

AUG 2023 – DEC 2023

- Created a customizable, WCAG 2 compliant website for an RPI club in HTML/CSS/Javascript.
- Planned all site design and led team members in their own implementations.

Project Lead – PSoft Tools

JAN 2025 – MAY 2025

- Led the creation of automatic practice problem generation and verification tools for the Principles of Software course. Developed the frontend in TypeScript/React/CSS and the backend in Python while using Node.js with Express for server-side functionality.
- Delegated team responsibilities, resolved team members' debug issues, maintained robust documentation, and coordinated development with past team members and faculty.

RPI Players

Sound Chair & Sound Designer

Troy, NY

AUG 2022 – MAY 2025

- Managed the club's audio inventory, valued at \$81k, including all purchases and budgeting.
- Designed and produced audio effects for student theater productions using DAW platforms.
- Onboarded new club members by providing training on audio equipment and software.

PROJECTS

Generative Composition Software – Personal Project

JUL 2024 – AUG 2024

- Developed a program generating musical compositions from image input. Coded in Python and Max/MSP, with a GUI built in CustomTkinter and UDP communication via python-osc.

Rainforest IV

Designer & Performer

Troy, NY

JAN 2024

- Designed and performed in the 51st anniversary of a transducer-based soundscape alongside its original designers. Coded in Max/MSP, operated devices with self-made TouchOSC controls.

HONORS

- Upsilon Pi Epsilon Honor Society Member (RPI) – RPI's chapter of an international computer science honor society.
- Alpha Psi Omega Honor Society Member (RPI) – RPI's chapter of a national theater honor society.